

DECODE LAB

INTERNSHIP REPORT

DATA ANALYTICS

PROJECT SUMMARY REPORT

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Project 1: Data Cleaning & Preparation

Objective

To clean a raw dataset by identifying missing values, checking duplicates, and validating data formats.

Tools Used

- Microsoft Excel

Tasks Performed

1. Identified missing values in CouponCode.
2. Replaced missing values with "No Coupon".
3. Checked for duplicate IDs.
4. Checked for duplicate rows.
5. Validated date formats.
6. Reviewed numeric and text data formats.
7. Prepared the cleaned dataset.

Results

Check	Result
Missing Values	0
Duplicate Rows	0
Duplicate IDs	0
Invalid Dates	0

Conclusion

The dataset was successfully cleaned and validated according to the project requirements.

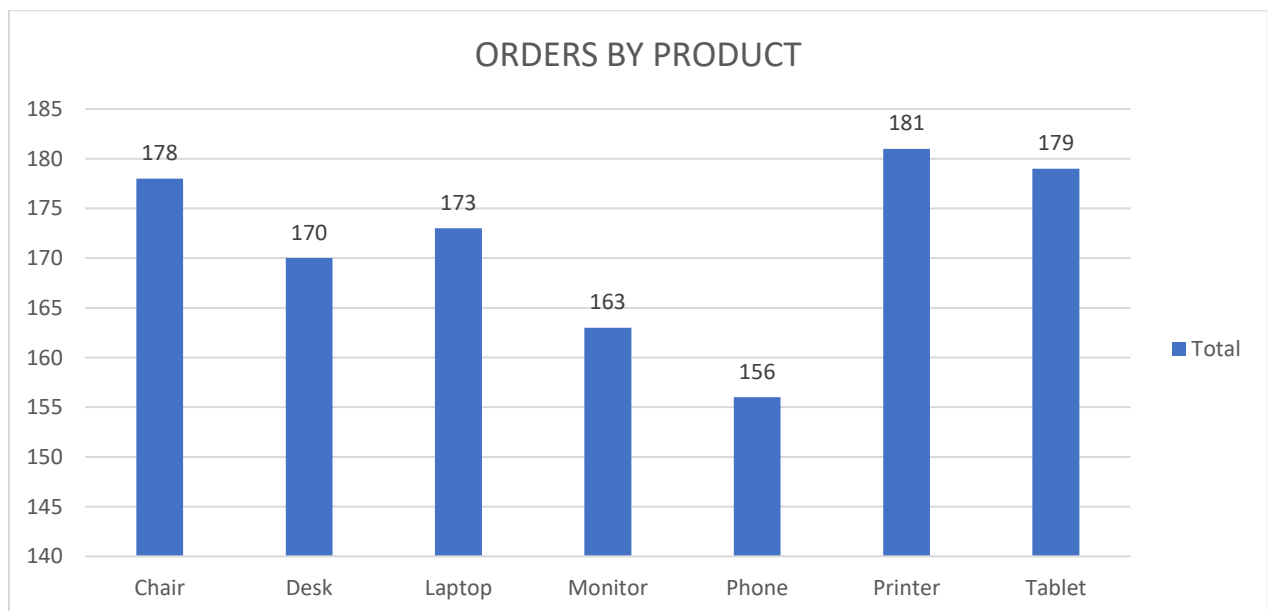
PROJECT 2

EDA ANALYSIS

PRODUCT ANALYSIS

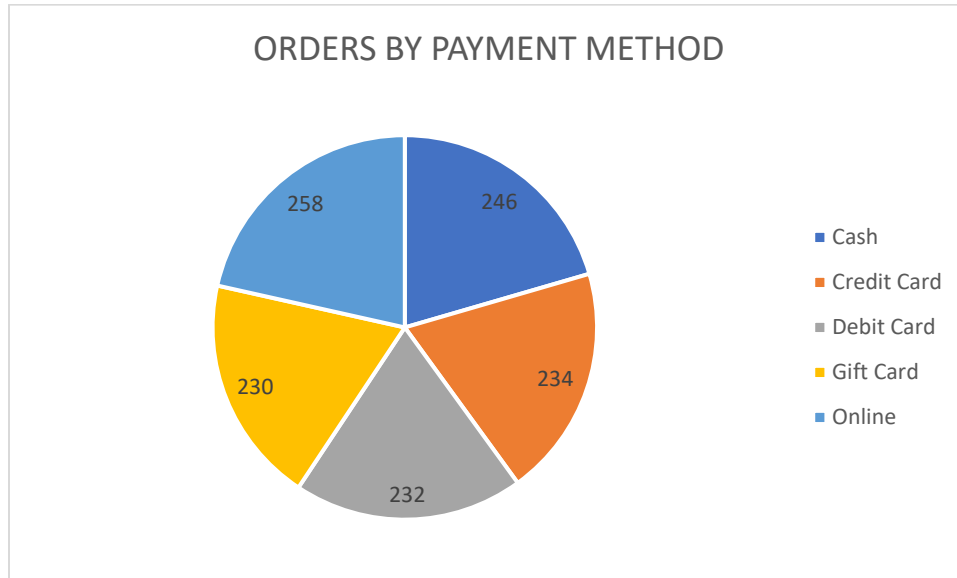
OBSERVATIONS:

- Printer has the highest number of orders (181).
- Phone has the lowest number of orders (156).
- The demand is fairly balanced across all products, with each product receiving between 156 and 181 orders.



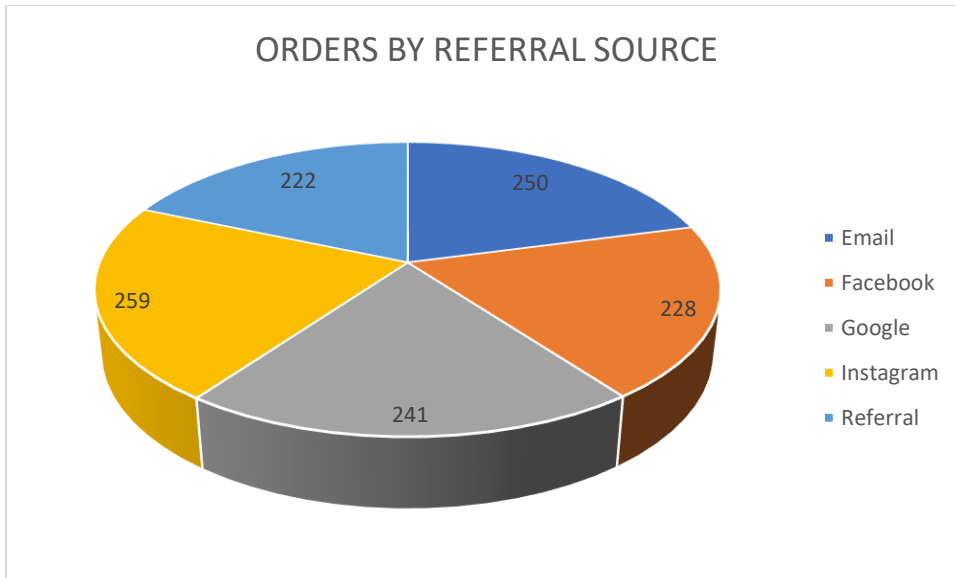
PAYMENT METHOD ANALYSIS:

- Online is the most preferred payment method with 258 orders.
- Gift Card is the least used payment method with 230 orders.
- Customer payment preferences are fairly balanced, with all payment methods having similar usage.



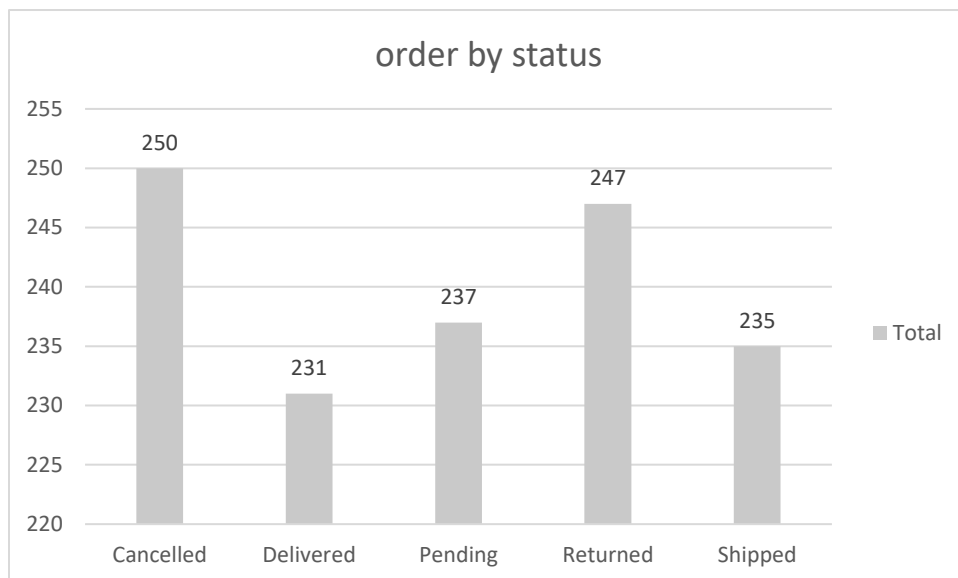
REFERRAL SOURCE ANALYSIS:

- The highest number of customers came from **INSTAGRAM**.
- The lowest number of customers came from **REFERRAL**.
- Customer acquisition is spread across multiple marketing channels, with no single channel dominating.



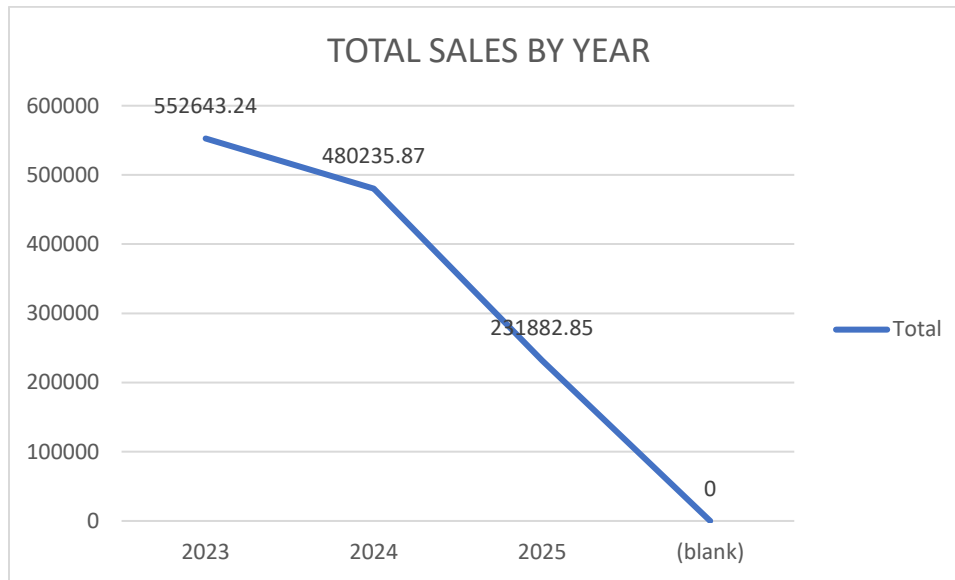
ORDER STATUS ANALYSIS:

- Cancelled orders are the highest with 250 orders.
- Delivered orders are the lowest with 231 orders.
- The distribution of order statuses is fairly balanced, indicating similar frequencies across all statuses.



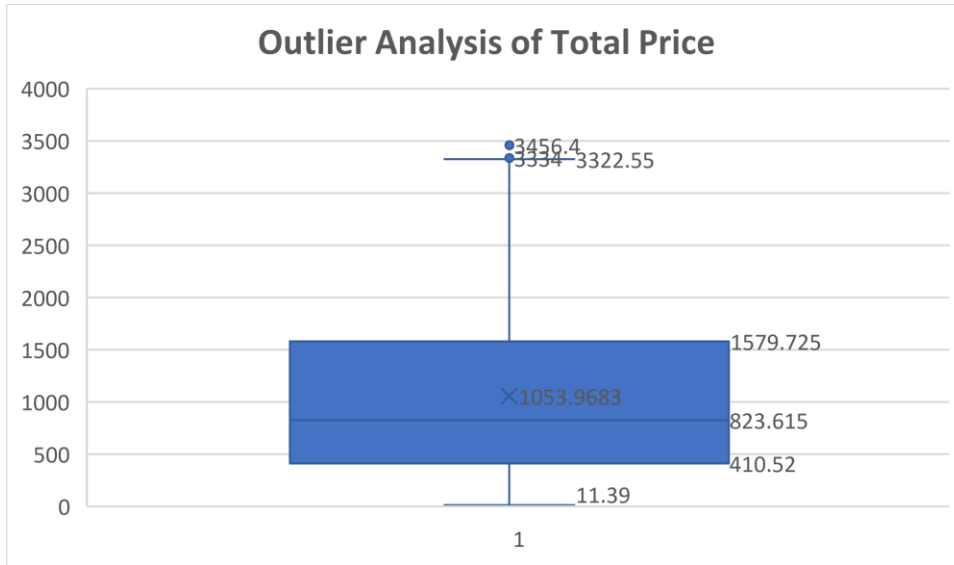
SALES TREND ANALYSIS:

- 2023 recorded the highest total sales (552,643.24).
- 2024 recorded total sales of 480,235.87.
- 2025 recorded the lowest total sales (231,882.85).
- Overall, total sales show a decreasing trend across the three years in this dataset.



OUTLIER ANALYSIS:

The Box & Whisker plot identified two high-value outliers in the TotalPrice data. These orders are significantly higher than the majority of transactions, indicating a small number of unusually expensive purchases.



Conclusion

This project provided a practical understanding of how data analysis can be performed using Microsoft Excel. By analyzing the e-commerce sales dataset, I was able to identify customer purchasing patterns, popular products, preferred payment methods, referral sources, and overall sales trends. Creating pivot tables, charts, and performing outlier analysis helped transform raw data into meaningful insights.

Through this project, I also improved my Excel skills, including data cleaning, descriptive statistics, pivot tables, pivot charts, and data visualization. Overall, this analysis demonstrates how businesses can use data to understand customer behavior, monitor sales performance, and make informed decisions to improve their operations.